



## Biomass development in slow sand filters

L.C. Campos, M.F.J. Su, N.J.D. Graham, S.R. Smith\*

*Department of Civil and Environmental Engineering, Imperial College of Science, Technology and Medicine, London SW7 2BU, UK*

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### Abstract

Microbial biomass development in the sand and schmutzdecke layer was determined in two full-scale slow sand filters, operated with and without a light excluding cover. A standard chloroform fumigation-extraction technique was adapted to routinely measure microbial biomass concentrations in the sand beds. Sand was sampled to a depth of 10 cm and schmutzdecke was also collected at the same random positions on the uncovered filter. Interstitial microbial biomass in the uncovered sand bed increased with time and decreased with sampling depth. There was a small accumulation of sand biomass with time in the covered filter, but no relationship was apparent between biomass concentration and depth in this filter. Schmutzdecke did not develop on the covered filter and was spatially highly variable in the uncovered condition compared to the consistent patterns observed in interstitial biomass production. It is speculated that microbial biomass in the sand of uncovered filters is largely related to carbon inputs from photosynthetic activity in the schmutzdecke and involves mechanisms that spatially distribute carbon substrate from the schmutzdecke to the sand. However, total organic carbon and dissolved organic carbon removals were similar in both filters suggesting that relatively small biomass populations in covered filters are sufficient to remove residual labile carbon during advanced water treatment and little further advantage to water purification and organic carbon removal is gained by the increased production of biomass in uncovered slow sand filter beds. © 2002 Elsevier Science Ltd. All rights reserved.

**Keywords:** Chloroform fumigation-extraction; Microbial biomass; Schmutzdecke; Slow sand filter; Water treatment

### 1. Introduction

Slow sand filtration (SSF) is one of the earliest forms of potable water treatment and remains an important process for water purification throughout the world. In recent years, a resurgence of interest in SSF has occurred, notably because of its efficiency at removing cysts of *Giardia* and *Cryptosporidium* enteroparasites [1–3], as well as dissolved organic matter after pre-oxidation [4]. In conventional water treatment works, SSF is generally the third stage of water purification after reservoir storage and rapid filtration, prior to disinfection [5]. However, slow sand filters can also provide a single-stage treatment for raw waters within certain water quality limits of turbidity and algal content [6]. Simplicity, and low capital and operating costs are

other principal advantages of SSF compared with more sophisticated methods of water treatment.

The high efficiency of water treatment achieved by slow sand filters is partly explained by the slow filtration rate ( $0.1\text{--}0.3\text{ m h}^{-1}$ ) and fine effective size of the sand ( $0.1\text{--}0.3\text{ mm}$ ), but is also attributed to biological processes in the layer of slime material that accumulates above the sand surface (schmutzdecke) and within the upper layers of the sand bed [7]. The schmutzdecke is also believed to be primarily responsible for the progressive increase of head loss observed during filter operation [1]. Despite its current and historical importance as a water treatment process, the fundamental physical and biological mechanisms controlling water purification and head loss development during SSF remain poorly defined.

Microbial biomass in the schmutzdecke and filter sand bed has been quantified using a range of different microbiological methods [8–10]. In some cases the sand

\*Corresponding author.

E-mail address: s.r.smith@ic.ac.uk (S.R. Smith).

biomass is determined indirectly by plate counts and numerical conversions [8]. The different approaches used to measure biomass concentrations and the inconsistent units and sampling intervals adopted (e.g. [11,10]) confound inter-study comparisons of biomass development and behaviour during SSF. Most reports are of single measurements of the net biomass production at the end of a filter run, prior to cleaning, and these emphasise the significant variability apparent in schmutzdecke and sand biomass accumulation in operational slow sand filters [11,10]. The collection of representative samples of schmutzdecke and sand material during filter operation is difficult in practice, and the lack of a simple routine method for measuring microbial biomass are probable reasons for the limited amount of field-scale investigation of the biological mechanisms of SSF. Detailed analyses of biomass growth in the schmutzdecke and within the sand bed during filter operation would improve understanding of the complex and fundamental interactions between the biological and physico-chemical processes operating in slow sand filter systems. This information would enable the development of mechanistic models of SSF systems which may improve the operational management of slow sand filters through the prediction of head loss rate, and the frequency of sand cleaning and renewal.

In this paper, we report the temporal and spatial dynamics of schmutzdecke and sand biomass development in full-scale slow sand filter beds at the Walton Water Treatment Works, operated by Thames Water Utilities Ltd., in the UK. This experimental investigation forms an integral part of ongoing studies at Imperial College, London, to develop a simulation model of SSF, incorporating biological and physico-chemical phenomena, to predict water filtration performance.

## 2. Materials and methods

In this paper, we define the schmutzdecke layer as the green algal mat formed on the top of the sand bed of the uncovered filter. The term 'schmutzdecke biomass' refers to the oven dry weight of the algal mat per unit area of sand bed, and includes all living organisms present in the slime layer. We define 'sand biomass' as the total microbial biomass concentration in the sand for both the uncovered and covered filters measured by a modified chloroform fumigation-extraction technique [12,13].

Sand and schmutzdecke samples were collected from two slow sand filter beds at the Walton Water Treatment Works operated by Thames Water Utilities Ltd. Walton is an advanced water treatment facility that receives raw water from the river Thames via large-scale storage

reservoirs. The reservoir water is treated sequentially by pre-ozonation, dissolved-air flotation, rapid filtration, intermediate ozonation, and granular activated carbon prior to SSF. The works treats between  $90 \times 10^3$  and  $137 \times 10^3$  m<sup>3</sup> of water per day and there are 9 slow sand filter beds in operation, each with an approximate area of 3500 m<sup>2</sup>.

Two dimensionally identical slow sand filter beds, receiving water from the same pre-treatment processes, were examined in this investigation (Table 1). Filter Bed 9 represented normal operational practice and was uncovered permitting full light interception. A second filter, Bed 10, was covered by a flexible plastic membrane, supported by positive air pressure, to exclude light as part of a wider investigation of SSF management. Filter Bed 10 was completely renewed with clean sand before the experimental period commenced and came into operation on the 11 May 2000. The sand surface of Bed 9 was skimmed following routine cleaning and water filtration began on 17 May 2000. Water temperature was measured during the experimental period and increased from initial values of 12–13°C in May 2000 to maxima of 18–19°C recorded after 100 days in August (Fig. 1).

Sand cores were collected whilst the filters were operating from a small boat, at intervals of approximately 2 weeks during the period 23 May–22 August 2000. Samples were collected using a specialised coring device developed by the Department of Geology, University College London [14]. The filter beds were divided into 140 sampling units with an approximate area of 5 m × 5 m. Three replicate cores of sand were sampled to a depth of 10 cm from randomly selected locations on each sampling event.

The thickness of the green algal mat forming the schmutzdecke was measured using a graduated depth gauge. The schmutzdecke and the supernatant water were separated from the sand cores collected from Bed 9

Table 1  
Characteristics of the filter beds

|                                      | Bed 9<br>(uncovered) | Bed 10<br>(covered) |
|--------------------------------------|----------------------|---------------------|
| Run start date                       | 17 May<br>2000       | 11 May<br>2000      |
| Run finish date                      | 28 August<br>2000    | <sup>a</sup>        |
| Duration of the run (d)              | 104                  | <sup>a</sup>        |
| Bed area (m <sup>2</sup> )           | 3500                 | 3500                |
| Filtration rate (m h <sup>-1</sup> ) | 0.15                 | 0.15                |
| Sand depth (m)                       | 0.8                  | 0.8                 |
| Effective size of sand (mm)          | 0.3                  | 0.3                 |
| Supernatant water depth (m)          | 1.5                  | 1.5                 |

<sup>a</sup> Run continued beyond study period.

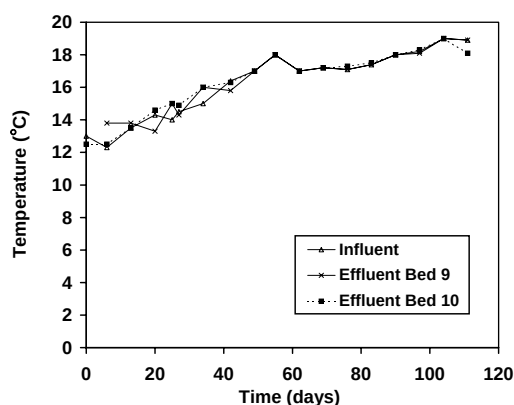


Fig. 1. Influent and effluent water temperature of uncovered (Bed 9) and covered (Bed 10) slow sand filters at Walton Treatment Works.

and placed in plastic bottles. Undisturbed sand cores were divided into five uniform segments of 2 cm, and the sub-samples from each section of the profile (0–2, 2–4, 4–6, 6–8, 8–10 cm) were pooled and thoroughly mixed to provide a representative composite for the determination of microbial biomass concentration in the sand. The pooled profile sections were transferred to plastic sample bags and both sand and schmutzdecke materials were transported to the laboratory in a cool box. The validity of pooling replicate sections of sand profiles was examined using a *t*-test statistic to compare the sand biomass content of a pooled sample with three independent measurements of the biomass concentration in individual cores collected on 25 July 2000 (day 69) at the same random locations on the filter bed. The *t* value was not statistically significant at  $P = 0.05$  indicating that the sand biomass measured in the composite sample was representative of the mean concentration measured from the analysis of individual replicate samples of sand (Fig. 2).

The dry weight of schmutzdecke collected from the sand surface was measured by a membrane filtration procedure [15]. Schmutzdecke was washed with mains supplied water, and filtered through 11 µm pore size, cellulose fibre filter papers and dried in a forced air oven maintained at 60°C.

A chloroform fumigation-extraction technique was adapted from a standard method for microbial biomass determination in soil [12] to quantify biomass concentration in the sand from slow sand filters. Sub-samples of sand (25 g fresh weight) from the pooled profile sections were transferred into four 100 ml glass bottles with plastic screw caps. A small beaker containing 30 ml of ethanol-free, amylene-stabilised chloroform (Rathburns) with 3–4 anti-bumping granules was placed in the base of a large capacity desiccator vessel fitted with a

vacuum tap and lined with moist absorbent paper to prevent sample desiccation. Two of the bottles from the set of four, for each composite sample, were placed on a supporting plate in the desiccator and up to 10 duplicate bottles could be simultaneously fumigated by this method. A high vacuum was applied to the desiccator for approximately 5 min until the chloroform had begun to boil. The desiccator was isolated from the vacuum after closing the gas-tight valve and incubated at room temperature in the dark for 24 hours. The vacuum was released to remove the chloroform and lining material, and the desiccator was cleaned and dried with absorbent laboratory paper. The sample bottles were replaced and the desiccator was evacuated under high vacuum for 3 min and ventilated. This procedure was repeated six times to remove residual chloroform from the samples. The unfumigated bottles were stored in the dark at 4°C or were extracted immediately for the determination of total organic carbon (TOC) content. Fumigated and unfumigated samples were extracted for 30 min in the sealed glass bottles with 50 ml of 0.5 M potassium sulphate solution using a laboratory shaker set at 200 revolutions  $\text{min}^{-1}$ . The TOC concentration in extracts of fumigated and unfumigated samples was measured using a Dohrmann DC80 automatic carbon analyser. The mean TOC concentration determined in the extracts of the unfumigated samples was subtracted from the mean result obtained for the fumigated sand to estimate the extractable organic carbon (OC) fraction released from the lysis of microbial cells by chloroform fumigation. This value was converted to biomass carbon using a multiplication factor of 2.22, following the approach recommended by Wu et al. [12].

Sand was collected from the filter beds in a water-saturated condition and this can interfere with the efficiency of chloroform fumigation [16]. Therefore, the moisture content of the sand was reduced to 40–50% of its water holding capacity (WHC) to aid chloroform penetration by absorption into laboratory filter paper. Water-saturated sand (100 g) was uniformly distributed on to double layers of 11 µm pore size, cellulose fibre filter paper (14 × 14 cm) and moisture from the sample was absorbed by the paper for a period of 10 min. This procedure was repeated with fresh filter paper and the WHC and gravimetric moisture content of the dewatered sand were determined at each step [17]. The moisture content of water-adjusted sand used for chloroform fumigation was reduced on average to 48% of the WHC.

The TOC and dissolved organic carbon (DOC) concentrations in influent and treated water from the slow sand filter beds are monitored routinely at Walton and these water quality data were provided by Thames Water for Bed 9 and Bed 10 for the duration of the experimental monitoring period.

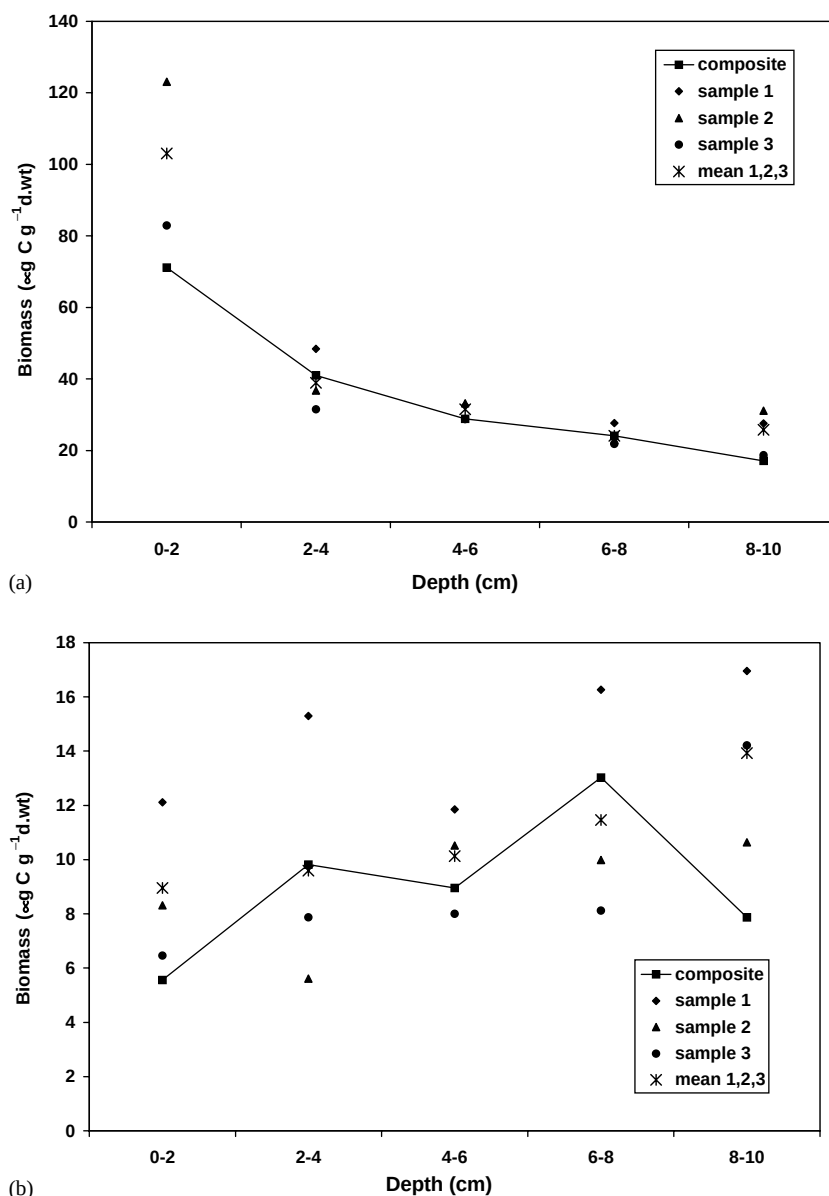


Fig. 2. Assessment of the spatial variation in sand biomass concentration on day 69 for (a) uncovered (Bed 9) and (b) covered (Bed 10) slow sand filters. The composite sample was produced from three replicate sand cores; samples 1–3 were individual replicate cores from the same sampling positions used to produce the composite. (Note: only duplicate measurements were available from the 0–2 cm layer for Bed 9).

### 3. Results and discussion

#### 3.1. Sand biomass

The general patterns in interstitial microbial biomass concentrations in the sand in relation to time and depth for both filters are presented in Fig. 3. In the uncovered system, sand biomass increased with time and decreased

with depth (Fig. 3a). The biomass carbon concentration measured in the top 0–2 cm of sand from the uncovered filter (Fig. 3a), with full light interception, increased generally to a maximum value of  $60 \mu\text{g C g}^{-1}$  (dry sand) in relation to filter run time, although more than  $120 \mu\text{g C g}^{-1}$  was recorded in the surface sand at day 56. Biomass accumulated at a slower rate in the covered sand compared to the uncovered filter and increased to a

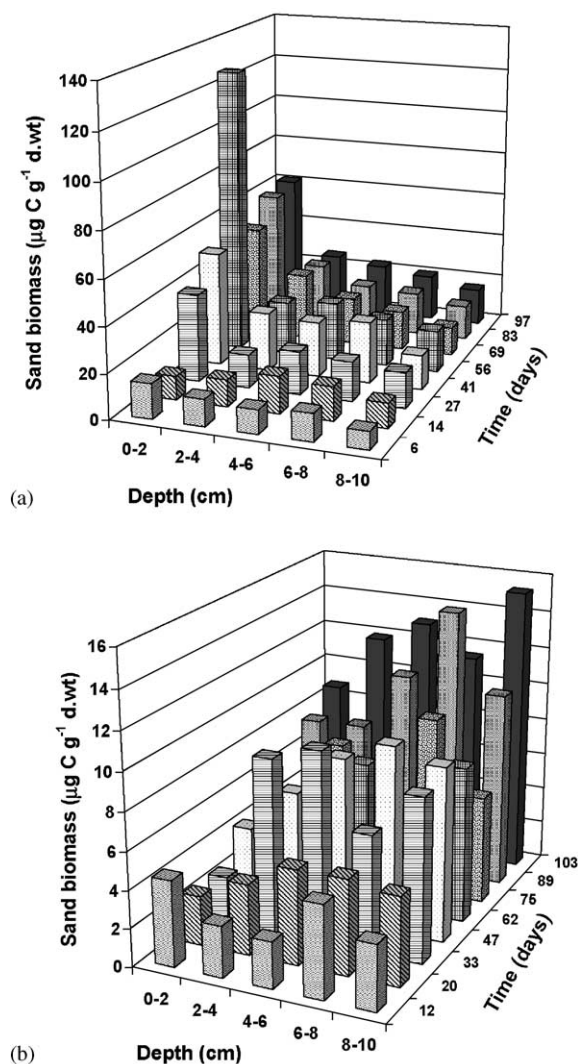


Fig. 3. Total microbial biomass C concentration in sand from (a) uncovered (Bed 9) and (b) covered (Bed 10) slow sand filters in relation to time and depth.

maximum concentration of  $15 \mu\text{g C g}^{-1}$  (dry sand) after 103 days (Fig. 3b). No consistent relationship was observed between biomass concentration and depth of sampling in the covered filter.

Temporal trends in the weighted average biomass concentrations in the top 10 cm of the sand beds are shown in Fig. 4. The biomass concentration increased significantly with time in the uncovered filter and was summarised by a simple logistic growth function ( $r^2 = 0.98$ ;  $P < 0.001$ ). The model predicted a maximum biomass accumulation in the sand of  $22.5 \mu\text{g C g}^{-1}$  (dry sand) after 97 days of operation and the filter run ended 7 days later when the terminal head loss was reached. These observations are consistent with other reports

[8,18] linking changes in head loss to biomass development in operational slow sand filters. Biomass in the covered sand bed also increased (Fig. 4), but at a significantly slower rate compared to the uncovered filter and in approximately linear relation with time ( $r^2 = 0.70$ ;  $P = 0.009$ ). The covered filter had not reached terminal head loss at the end of the monitoring period (day 103) and continued to operate normally after the final sample collection event in August 2000. Unfortunately, head loss is not routinely measured on the slow sand filters at Walton and so the relationship between biomass growth and head loss development could not be further elucidated.

The distribution of interstitial biomass with depth in sand from the uncovered and covered filters is shown in Fig. 5. No measurable increase in the microbial biomass content of the upper 0–2 cm sand layer was detected for the first 14 days of operation of the uncovered filter and the correlation between biomass content and depth achieved statistical significance after 56 days of operation for this filter. Similar patterns in biomass distribution with depth in slow sand filters have also been reported by Duncan [8] and Yordanov et al. [10]. As expected, the largest accumulation of biomass in the uncovered filter was measured in the top 0–2 cm of sand. The increased production of biomass in the uncovered filter emphasised the importance of photosynthetic inputs of carbon substrates to the sand from the schmutzdecke, compared to the covered sand bed, where growth was smaller and supported only by the availability of substrates supplied in the influent water (Fig. 6). Nevertheless, the observed reduction in biomass concentration with depth suggested there was also substrate limitation to microbial growth in the deeper layers of the uncovered filter.

The biologically active microbial community in the top few centimetres of sand in the filter bed is considered to be a major factor contributing to water purification by SSF [1,19]. Interestingly, no differences were apparent in the water quality of effluents from either filter and generally consistent removals of TOC and DOC were observed throughout (Fig. 6), despite large differences in the biological properties of the covered and uncovered filter beds. Thus, the average removals of TOC and DOC by Bed 9 (uncovered) were 25% and 23%, respectively, and Bed 10 removed 23% of both TOC and DOC on average from the influent water (note: mean values were estimated from net positive removal data for TOC/DOC presented in Fig. 6). This behaviour contrasts with the data reported by Collins et al. [20], which showed increased removals of natural organic matter (NOM) and organic precursor material in covered filters with increasing biomass concentration in the sand. These apparently conflicting observations could be explained by differences in the quality of influent water, and thus the nature of the dissolved

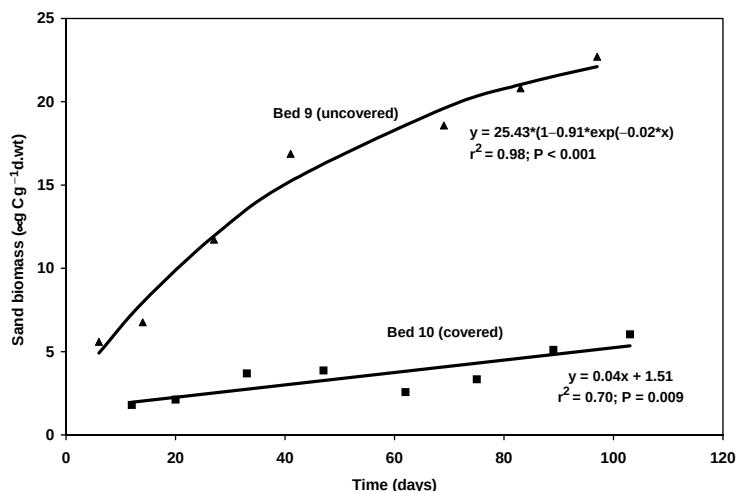
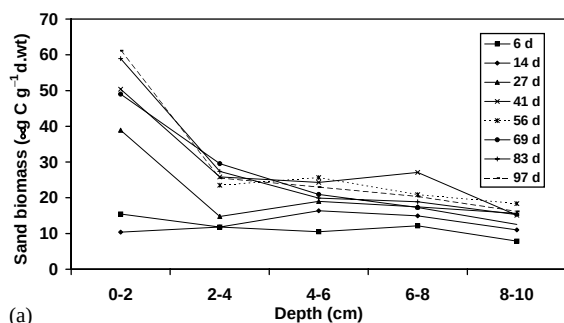
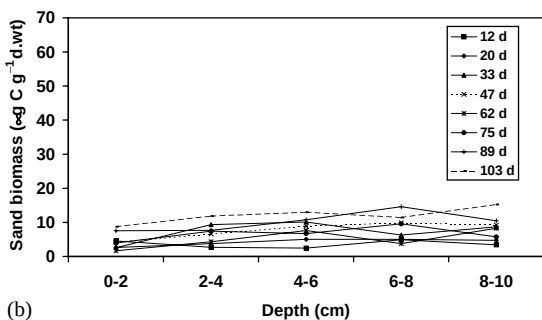


Fig. 4. Weighted average biomass C concentrations in the 0–10 cm sand layer from slow sand filters in relation to time (biomass value at 0–2 cm depth in Bed 9 for day 56 was  $126.8 \mu\text{g C g}^{-1} \text{d.wt}$  and the weighted average result was omitted from the regression).



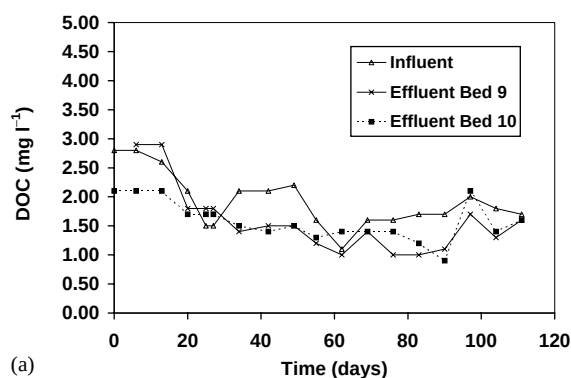
(a)



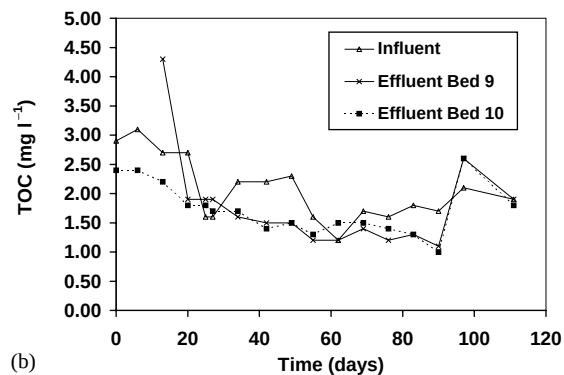
(b)

Fig. 5. Interstitial sand biomass concentration in relation to depth in (a) uncovered (Bed 9), and (b) covered (Bed 10) slow sand filters.

organics, supplied to the filters. In our study, SSF was part of an advanced water treatment system involving pre-treatment by reservoir, pre-ozonation, flotation, rapid filtration, intermediate ozonation, and granular activated carbon. In view of this, it is likely that the amount of bio-available organic substrates in the influent to the filters at Walton would be very low. However, Collins et al. [20] conducted their studies using



(a)



(b)

Fig. 6. Influent and effluent (a) DOC and (b) TOC concentrations for uncovered (Bed 9) and covered (Bed 10) filters. (Data provided by Thames Water Utilities Ltd.).

a more conventional SSF system, without extensive pre-treatment of the influent water, and therefore it is likely that a greater amount of bio-available organic material was present.

### 3.2. Schmutzdecke thickness and biomass

Schmutzdecke was visible on the surface of the sand of Bed 9 from day 14 onwards. However, samples could not be collected until day 41 because the schmutzdecke had not developed sufficiently for it to be physically separated as a distinct layer from the sand before this period of filter operation had elapsed. By day 41, schmutzdecke had also formed on most of the filter surface and at the random locations used for collecting sand cores, increasing the possibility of sample collection.

Biomass accumulation in the schmutzdecke layer was highly variable and showed no consistent patterns of spatial or temporal development (Figs. 7 and 8). The substantial spatial variability of the schmutzdecke contrasts with the relatively uniform patterns of microbial biomass growth observed in the sand. Despite this inconsistency it is speculated that the schmutzdecke is responsible for significant carbon inputs to the underlying sand layer. Bellinger [11] estimated that up to 30% of the synthesised organic matter produced by algae in the supernatant water and surface of the sand is passed into the filter bed, and is consumed by bacteria. Carbon distribution to the sand may be the product of both physico-chemical diffusion as well as biological

processes including, for example, the activities of midge larvae [21].

The thickness of the schmutzdecke was approximately 1 mm after 27 days of filter operation and increased to 3 mm after 41 days at the random sampling locations on the filter bed. However, in other areas of the filter it was apparent that much thicker layers (30–50 mm) of the

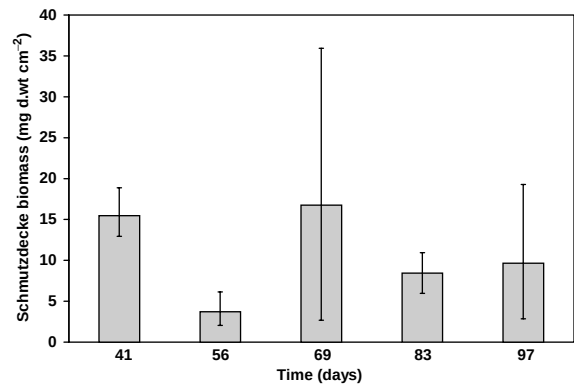


Fig. 8. Temporal variation in mean schmutzdecke biomass in the uncovered filter (vertical bars represent minimum and maximum values).

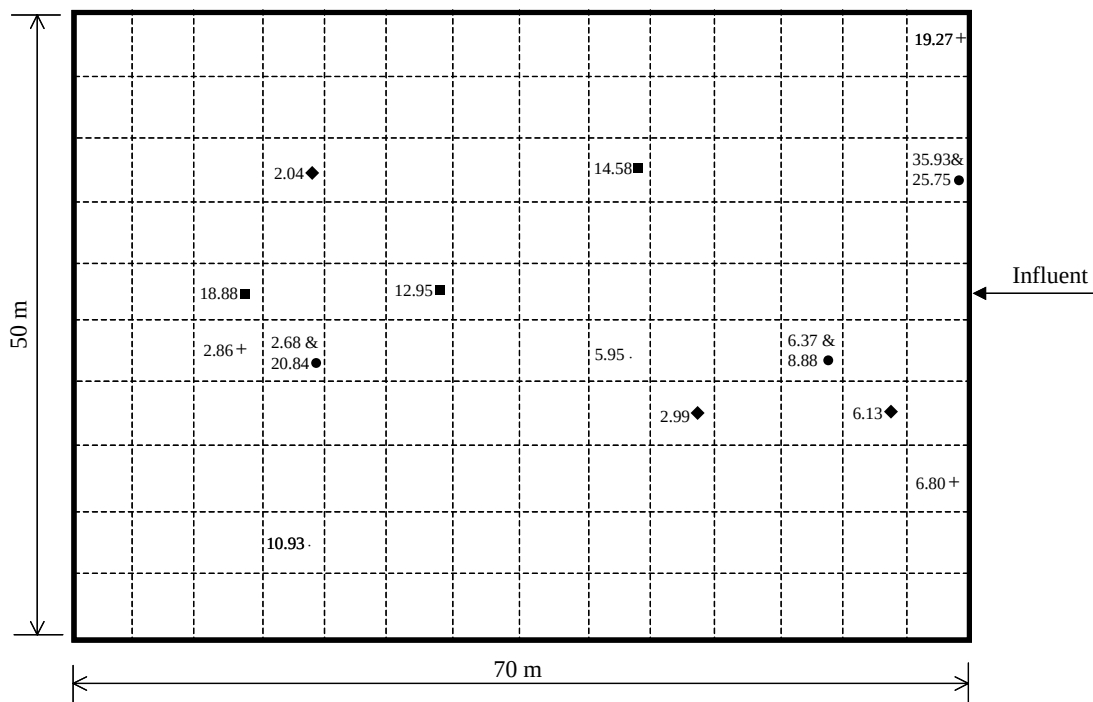


Fig. 7. Spatial and temporal distribution of the schmutzdecke biomass in the uncovered filter ( $\text{mg d.wt cm}^{-2}$ ). Sample time (days): (■) 41, (◆) 56, (●) 69, (×) 83 (+) 97 (note: grid squares represent the sampling units of approximately  $5 \text{ m} \times 5 \text{ m}$ ).

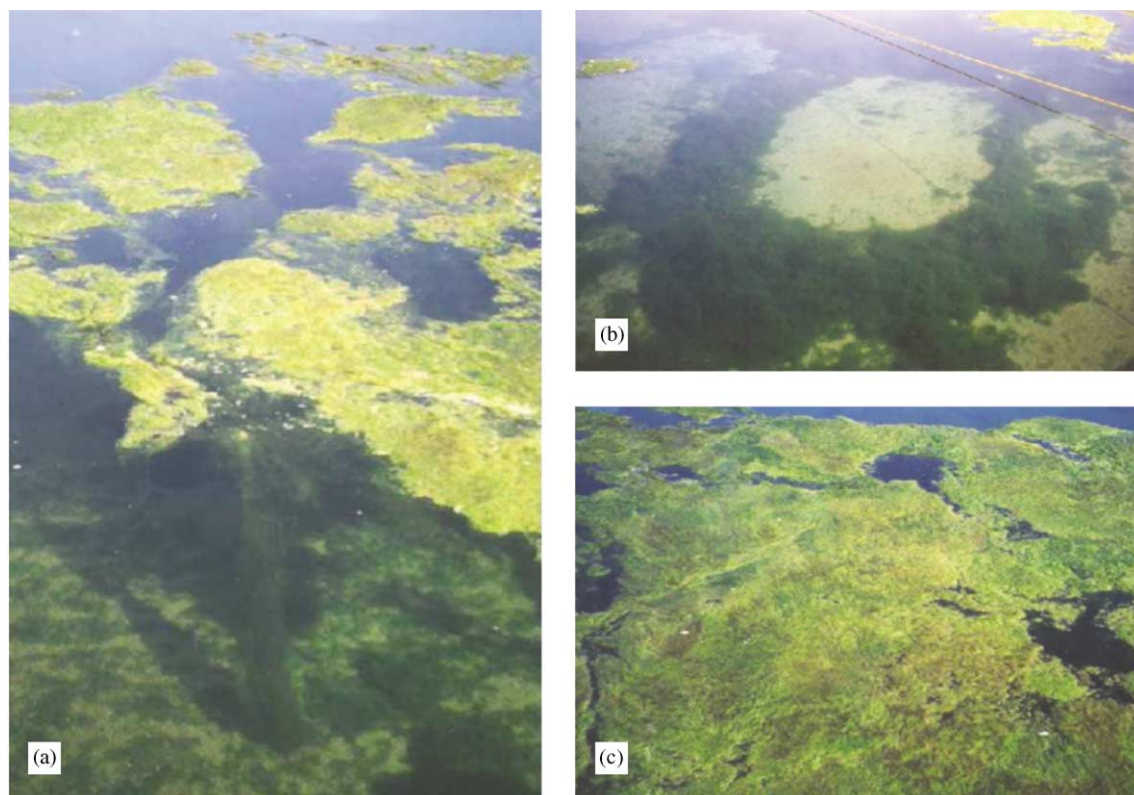


Plate 1. Algal development in the uncovered slow sand filter (Bed 9) at Walton Works: (a) supernatant algae; (b) benthic algae and (c) algal mat on the water surface.

schmutzdecke had developed. After 56 days, the thickness of the schmutzdecke could not be measured because the proliferation of filamentous algae in the supernatant water interfered with the sampling procedure for benthic matter and disturbed the schmutzdecke layer. The extent of benthic and supernatant algal development in Bed 9 at Walton Water Treatment Works after 69 days of operation is shown in Plate 1. Although species identification was not an objective of this investigation, similar prolific algal development in uncovered slow sand filters has been observed by others [22–24,11].

#### 4. Conclusions

Chloroform fumigation-extraction provides a simple and routine method for the determination of microbial biomass in slow sand filter beds for potable water treatment. Interstitial microbial biomass in the uncovered filter studied here increased generally with time and decreased with depth, and most biomass development occurred in the upper 0–2 cm layer of the filter. By contrast, biomass content was much lower in the

covered bed and only a modest increase was observed with time during the monitoring period, but no relationship was apparent with depth.

Covered and uncovered slow sand filters may achieve similar rates of OC removal in advanced water treatment systems, despite having markedly different biological properties. The results presented have suggested that, in advanced water treatment systems, labile OC is effectively removed and utilised for growth by the microbial biomass in covered slow sand filters. Uncovered filters, on the other hand, support much more prolific biomass development, but this has no apparent benefit for OC removal. Indeed, biomass accumulation may be disadvantageous in advanced water treatment systems as it is linked to head loss development [8,18] and terminal head loss occurs much earlier in uncovered filters compared to covered systems where sunlight is excluded.

The schmutzdecke layer in the uncovered filter was spatially highly variable and showed no consistent pattern of development in relation to sampling time. This contrasted with the relatively uniform spatial accumulation of interstitial microbial biomass in the sand. Nevertheless, it is suggested that the schmutzdecke



is responsible for significant inputs of carbon substrates to the underlying sand layer supporting interstitial microbial growth, possibly distributed by diffusion and biological processes (e.g. activities of midge larvae). Further studies are required to verify this and also to study the dynamic nature of the benthic and supernatant algal development and movement on the sand.

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